Lab 2: Data Preprocessing

Các nội dung chính

1. Mục tiêu:

- Nắm được các bước cơ bản trong khâu tiền xử lí dữ liệu.

2. Dữ liệu:

- Dữ liệu bất động sản - *Bengaluru_House_Data*

> Gồm các trường dữ liệu: location, size, total_sqft, price, ...

> Link Kaggle: https://www.kaggle.com/amitabhajoy/bengaluru-house-price-data

3. Yêu cầu:

- Sử dụng các công cụ (Pandas, Seaborn, ...) để thực hiện xem xét, đánh giá đặc điểm của dữ liệu, từ đó đưa ra phương án tiền xử lí dữ liệu (làm sạch, trích xuất thông tin ban đầu, ...)

Nội dung thực hành

1. Chuẩn bị các thư viện cần thiết

In [1]:
```
import numpy as np #Làm việc với các dữ liệu mảng nhiều chiều
import pandas as pd #Giúp làm việc với các dữ liệu dạng bảng
import matplotlib.pyplot as plt #Thư viện hỗ trợ trực quan hóa dữ liệu
import seaborn as sns #Thư viện giúp trực quan hóa dữ liệu, được xây trên matplotli
```

2. Load dữ liệu từ file đã tải về

2.1. Đọc dữ liệu bằng pandas, dạng dataframe

In [2]:
```
path = "./Lab 2-Data Preprocessing/Bengaluru_House_Data.csv"
df_raw = pd.read_csv(path)
df_raw.shape
```

Out[2]:
```
(13320, 9)
```

2.2. Xem 5 mẫu đầu tiên

In [3]:
```
df_raw.head() # return DataFrame
```

Out[3]:

	area_type	availability	location	size	society	total_sqft	bath	balcony
0	Super built-up Area	19-Dec	Electronic City Phase II	2 BHK	Coomee	1056	2.0	1.0
1	Plot Area	Ready To Move	Chikka Tirupathi	4 Bedroom	Theanmp	2600	5.0	3.0
2	Built-up Area	Ready To Move	Uttarahalli	3 BHK	NaN	1440	2.0	3.0
3	Super built-up Area	Ready To Move	Lingadheeranahalli	3 BHK	Soiewre	1521	3.0	1.0
4	Super built-up Area	Ready To Move	Kothanur	2 BHK	NaN	1200	2.0	1.0

2.3. Xem 5 mẫu cuối cùng

In [4]:
```
df_raw.tail()
```

Out[4]:

	area_type	availability	location	size	society	total_sqft	bath	balcor
13315	Built-up Area	Ready To Move	Whitefield	5 Bedroom	ArsiaEx	3453	4.0	0
13316	Super built-up Area	Ready To Move	Richards Town	4 BHK	NaN	3600	5.0	Na
13317	Built-up Area	Ready To Move	Raja Rajeshwari Nagar	2 BHK	Mahla T	1141	2.0	1
13318	Super built-up Area	18-Jun	Padmanabhanagar	4 BHK	SollyCl	4689	4.0	1
13319	Super built-up Area	Ready To Move	Doddathoguru	1 BHK	NaN	550	1.0	1

3. Phân tích dữ liệu thăm dò (Exploratory Data Analysis-EDA)

In [5]:
```
df = df_raw.copy() #Tạo bản sao để thực hiện EDA
```

3.1. Thông tin cơ bản về dữ liệu: tên, số giá trị non-null, kiểu dữ liệu của từng trường

In [6]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 13320 entries, 0 to 13319
Data columns (total 9 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   area_type     13320 non-null  object
 1   availability  13320 non-null  object
 2   location      13319 non-null  object
 3   size          13304 non-null  object
 4   society       7818 non-null   object
 5   total_sqft    13320 non-null  object
 6   bath          13247 non-null  float64
 7   balcony       12711 non-null  float64
 8   price         13320 non-null  float64
dtypes: float64(3), object(6)
memory usage: 936.7+ KB
```

3.2. Thống kê một số thuộc tính cơ bản của dữ liệu: count, mean, std, min, max, quartile

In [7]:
```
df.describe()
```

Out[7]:

	bath	balcony	price
count	13247.000000	12711.000000	13320.000000
mean	2.692610	1.584376	112.565627
std	1.341458	0.817263	148.971674
min	1.000000	0.000000	8.000000
25%	2.000000	1.000000	50.000000
50%	2.000000	2.000000	72.000000
75%	3.000000	2.000000	120.000000
max	40.000000	3.000000	3600.000000

3.3. Thống kê các giá trị duy nhất của từng trường và số lần xuất hiện của chúng

In [8]:
```
def value_count(df):
  for var in df.columns:
    print(df[var].value_counts())
    print("--------------------------------")

value_count(df)
```

```
area_type
Super built-up  Area    8790
Built-up  Area          2418
Plot  Area              2025
Carpet  Area              87
Name: count, dtype: int64
--------------------------------
availability
Ready To Move    10581
18-Dec             307
18-May             295
18-Apr             271
18-Aug             200
                 ...
15-Aug               1
17-Jan               1
16-Nov               1
16-Jan               1
14-Jul               1
Name: count, Length: 81, dtype: int64
--------------------------------
location
Whitefield                        540
Sarjapur  Road                    399
Electronic City                   302
Kanakpura Road                    273
Thanisandra                       234
                                 ...
Bapuji Layout                       1
1st Stage Radha Krishna Layout      1
BEML Layout 5th stage               1
singapura paradise                  1
Abshot Layout                       1
Name: count, Length: 1305, dtype: int64
--------------------------------
size
2 BHK         5199
3 BHK         4310
4 Bedroom      826
4 BHK          591
3 Bedroom      547
1 BHK          538
2 Bedroom      329
5 Bedroom      297
6 Bedroom      191
1 Bedroom      105
8 Bedroom       84
7 Bedroom       83
5 BHK           59
9 Bedroom       46
6 BHK           30
7 BHK           17
1 RK            13
10 Bedroom      12
9 BHK            8
8 BHK            5
```

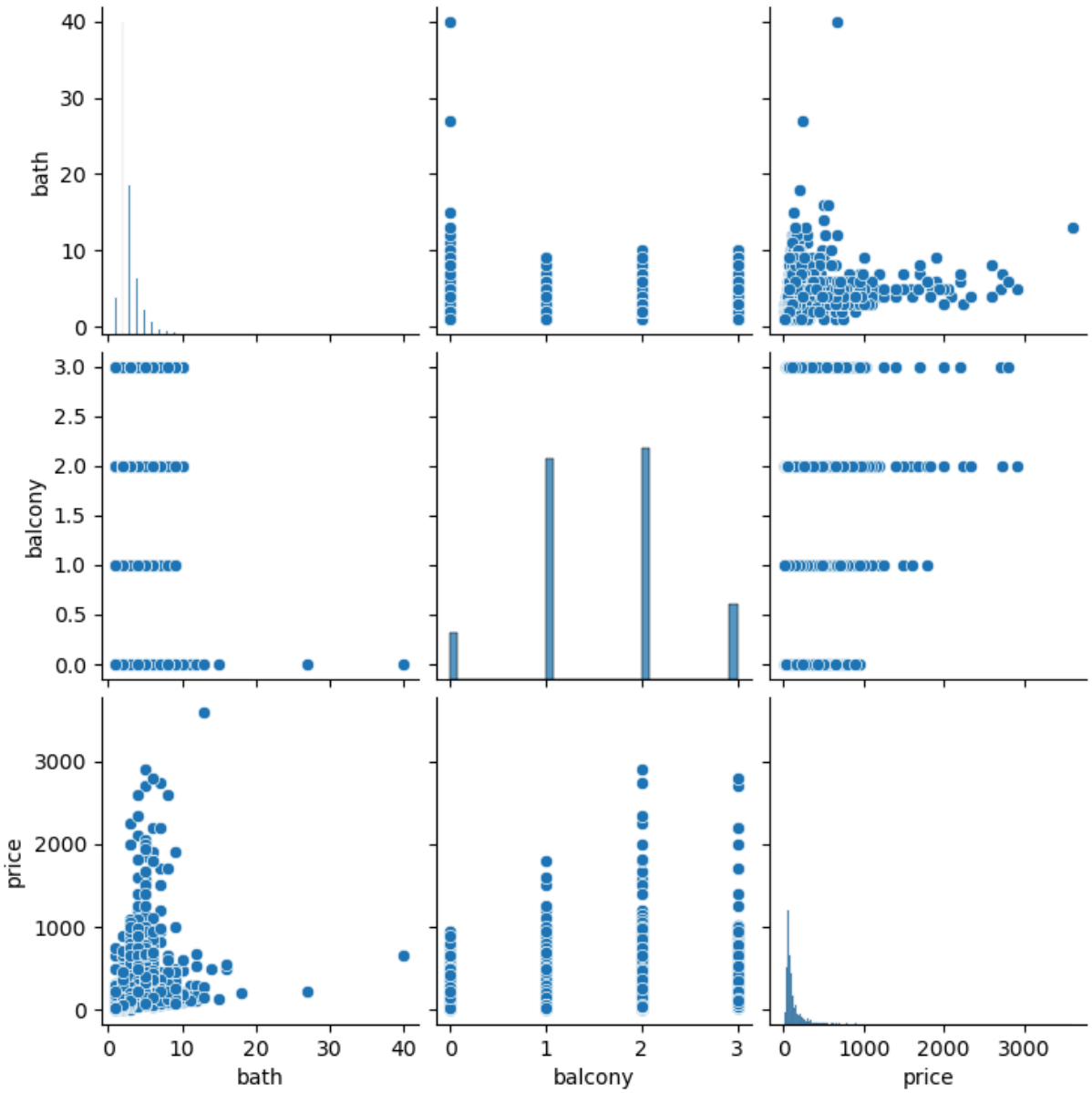
```
11 BHK           2
11 Bedroom       2
10 BHK           2
14 BHK           1
13 BHK           1
12 Bedroom       1
27 BHK           1
43 Bedroom       1
16 BHK           1
19 BHK           1
18 Bedroom       1
Name: count, dtype: int64
--------------------------------
society
GrrvaGr    80
PrarePa    76
Sryalan    59
Prtates    59
GMown E    56
           ..
Amionce     1
JaghtDe     1
Jauraht     1
Brity U     1
RSntsAp     1
Name: count, Length: 2688, dtype: int64
--------------------------------
total_sqft
1200    843
1100    221
1500    205
2400    196
600     180
       ...
3580      1
2461      1
1437      1
2155      1
4689      1
Name: count, Length: 2117, dtype: int64
--------------------------------
bath
2.0     6908
3.0     3286
4.0     1226
1.0      788
5.0      524
6.0      273
7.0      102
8.0       64
9.0       43
10.0      13
12.0       7
13.0       3
11.0       3
16.0       2
```

```
27.0       1
40.0       1
15.0       1
14.0       1
18.0       1
Name: count, dtype: int64
--------------------------------
balcony
2.0    5113
1.0    4897
3.0    1672
0.0    1029
Name: count, dtype: int64
--------------------------------
price
75.00     310
65.00     302
55.00     275
60.00     270
45.00     240
         ...
351.00      1
54.10       1
80.64       1
32.73       1
488.00      1
Name: count, Length: 1994, dtype: int64
--------------------------------
```

3.4. Xem xét tương quan về giá trị của các cặp trường số

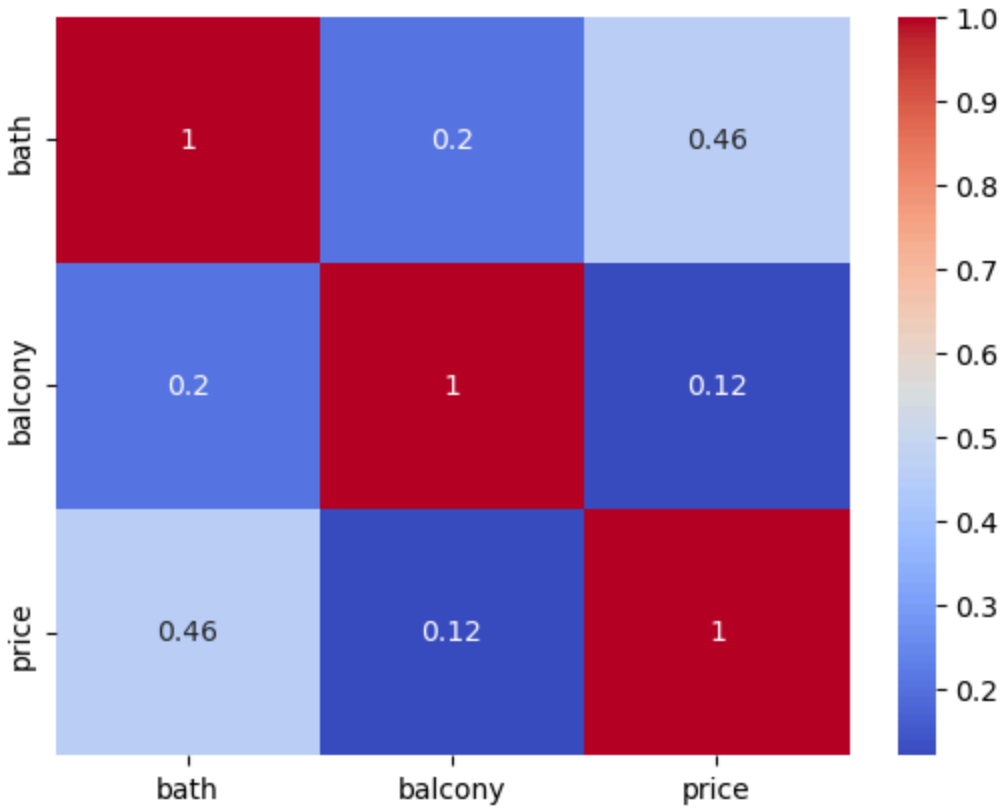
In [9]:
```
sns.pairplot(df)
```

Out[9]:
```
<seaborn.axisgrid.PairGrid at 0x18ef7a12660>
```



In [10]:
```
num_vars = ["bath", "balcony", "price"]
sns.heatmap(df[num_vars].corr(),cmap="coolwarm", annot=True)
```

Out[10]: <Axes: >



4. Tiền xử lý dữ liệu

4.1. Data Cleaning & Data Transformation

4.1.1. Xử lí các giá trị Null/ NaN

1. Thống kê tỉ lệ giá trị null của từng thuộc tính

In [11]:
```
df.isnull().mean()*100  # Tỷ lệ giá trị null của từng thuộc tính
```

Out[11]:
```
area_type        0.000000
availability     0.000000
location         0.007508
size             0.120120
society         41.306306
total_sqft       0.000000
bath             0.548048
balcony          4.572072
price            0.000000
dtype: float64
```

2. Loại đi trường society vì tỉ lệ null cao (41%)

In [12]:
```
df2 = df.drop('society', axis='columns')
df2.shape
```

Out[12]: (13320, 8)

3. Thay thế giá trị null trong trường balcony bằng giá trị trung bình của các giá trị non-null

In [13]:
```
df2['balcony'] = df2['balcony'].fillna(df2['balcony'].mean())
df2.isnull().sum()
```

Out[13]:
```
area_type        0
availability     0
location         1
size            16
total_sqft       0
bath            73
balcony          0
price            0
dtype: int64
```

4. Xóa đi các mẫu dữ liệu (hàng) có giá trị NaN (không có giá trị)

In [14]:
```
df3 = df2.dropna()
df3.shape
```

Out[14]: (13246, 8)

In [15]:
```
df3.isnull().sum() #Thống kê lại xem đã xử lí hết các dữ liệu null hay chưa?
```

Out[15]:
```
area_type       0
availability    0
location        0
size            0
total_sqft      0
bath            0
balcony         0
price           0
dtype: int64
```

4.1.2. Xử lí các trường thuộc tính

In [16]:
```
# Xem toàn bộ giá trị có thể có của dữ liệu
pd.set_option("display.max_columns", None)
pd.set_option("display.max_rows", None)
```

1. Converting 'total_sqft' cat feature in numeric (float)

In [17]:
```
# Xem xét trường total_sqft
df3['total_sqft'].value_counts()
```

Out[17]:

total_sqft	
1200	843
1100	221
1500	204
2400	195
600	180
1000	172
1350	132
1050	123
1300	117
1250	114
900	112
1400	108
1800	104
1150	101
1600	100
1140	91
2000	82
1450	70
1650	69
800	67
3000	66
1075	66
1020	63
2500	62
1125	60
1160	60
1550	60
950	59
1700	58
1180	58
1260	57
1255	56
1080	55
1220	55
1070	53
750	52
700	52
4000	48
1175	48
1225	48
1320	46
1240	46
2100	46
1230	45
1060	45
1210	44
850	43
1280	42
1270	41
1185	41
1190	40
1170	40
1410	40
1750	39
1025	38

1330	38
1310	37
1290	37
1850	37
1065	36
1194	36
1215	35
1090	35
500	34
1360	33
2700	33
1115	33
1900	32
3500	32
1120	32
1464	32
1530	31
2200	31
1205	31
1430	31
1340	31
1035	30
1165	30
1560	30
1130	29
1145	29
1128	29
3600	29
1355	28
1040	28
2800	28
1275	28
1105	27
1420	27
1155	27
1680	27
1760	25
1245	25
1216	25
1460	25
1590	25
650	25
1010	24
1475	24
1305	24
2600	24
1030	23
1110	23
1440	23
883	23
1575	22
1495	22
985	22
1246	22
1315	21
1610	21

1385	21
1027	21
1015	21
1470	21
1325	21
3200	21
1370	21
1243	21
660	20
1480	20
550	20
5000	20
925	20
1540	20
1390	20
1645	19
920	19
1365	19
1520	19
1265	19
1640	19
1570	19
1485	18
1095	18
4800	18
980	18
1525	18
1196	18
1345	18
1665	18
975	18
1195	18
1012	18
1820	18
2072	18
940	18
1375	17
960	17
400	17
1157	17
1295	17
1232	17
525	17
1045	16
2300	16
1830	16
1490	16
1418	16
1655	16
935	16
1135	16
3300	16
645	16
1197	16
1720	16
1235	16

1425	16
1153	15
1950	15
1660	15
1380	15
1116	15
4500	15
1740	15
1152	15
1085	15
1285	15
1920	15
1580	15
630	15
905	14
450	14
1630	14
1141	14
1445	14
720	14
675	14
1595	14
3900	13
3800	13
1875	13
1615	13
1082	13
1730	13
1535	13
840	13
1710	13
1690	13
1282	13
1510	13
1005	13
1314	13
1625	13
2215	13
970	13
2250	13
1033	12
1565	12
1404	12
1007	12
1296	12
2900	12
1151	12
1339	12
1691	12
1435	12
1890	12
1455	12
1056	12
1252	12
3100	12
1843	12

1162	11
1405	11
1174	11
1452	11
1515	11
845	11
1206	11
984	11
1804	11
1639	11
1346	11
540	11
1685	11
2150	11
965	11
1198	10
1256	10
3400	10
918	10
2350	10
1620	10
880	10
1415	10
995	10
1465	10
4200	10
1555	10
620	10
930	10
770	10
760	10
3596	10
1161	10
1063	10
957	10
1308	10
1810	10
1092	9
1605	9
1419	9
2180	9
780	9
1693	9
1224	9
2750	9
1715	9
910	9
1223	9
1724	9
6000	9
945	9
1186	9
1482	9
1725	9
3150	9
1113	9

1156	9
705	9
1047	9
3750	9
1756	9
890	9
2790	9
1790	9
1745	9
1870	9
2475	9
674	9
3250	9
825	9
1108	9
1222	9
5400	8
1133	8
875	8
1636	8
2990	8
1088	8
812	8
1335	8
1221	8
1178	8
1299	8
1705	8
1835	8
1146	8
656	8
1532	8
680	8
1395	8
1447	8
1352	8
1187	8
1163	8
1476	8
1026	8
1703	8
870	8
2145	8
1089	8
1322	8
990	8
1192	8
1670	8
2650	8
4400	8
3122	8
1276	7
1101	7
1139	7
1044	7
1166	7

1718	7
1183	7
1397	7
820	7
1093	7
1357	7
1112	7
1041	7
1096	7
1374	7
1545	7
1427	7
1306	7
1132	7
1262	7
1564	7
1840	7
1277	7
982	7
1272	7
1251	7
2050	7
1762	7
1254	7
1508	7
1199	7
708	7
1917	7
1910	7
1179	7
615	7
1880	7
1392	7
7500	7
1143	7
1847	7
1349	7
1069	7
1342	7
1036	7
1213	6
440	6
530	6
1278	6
1614	6
1846	6
2480	6
560	6
1787	6
1635	6
1173	6
625	6
924	6
1303	6
1362	6
1484	6

1697	6
1453	6
1084	6
1242	6
1053	6
1780	6
1735	6
710	6
7000	6
2040	6
1016	6
1583	6
605	6
2425	6
1411	6
1755	6
1304	6
1107	6
2010	6
1274	6
1675	6
933	6
1719	6
3895	6
1247	6
860	6
830	6
1541	6
1893	6
1785	6
1512	6
1167	6
520	6
654	6
1181	6
1451	6
2690	6
1257	6
610	6
1936	6
2275	6
1571	6
1856	6
1268	6
1313	6
936	6
2774	6
1127	6
1333	6
1372	6
1656	6
1176	6
1602	6
1770	6
1991	6
1286	6

1028	6
1717	6
1855	6
810	6
929	6
1083	6
1019	6
1062	6
1102	6
2280	6
2225	6
1126	6
1329	5
1865	5
1052	5
1328	5
919	5
1123	5
1021	5
2710	5
1263	5
1884	5
1344	5
4395	5
4050	5
1076	5
1754	5
1801	5
1505	5
545	5
1226	5
1293	5
1868	5
1664	5
1253	5
1168	5
1244	5
915	5
1403	5
1662	5
907	5
1297	5
2254	5
410	5
1826	5
1309	5
1559	5
2850	5
1267	5
1326	5
1798	5
1738	5
1057	5
1930	5
1654	5
1077	5

1009	5
1533	5
1862	5
1148	5
1573	5
420	5
2360	5
635	5
1586	5
2830 - 2882	5
1055	5
1022	5
865	5
1358	5
1408	5
1765	5
1935	5
510	5
1683	5
1852	5
1098	5
715	5
1825	5
1458	5
1184	5
1188	5
1208	5
1749	5
3450	5
1231	5
2119	5
740	5
1436	5
1204	5
1351	5
640	5
1142	5
1354	5
1925	5
1567	5
1364	5
1457	5
1444	5
1031	5
1585	5
1109	5
1269	5
2340	5
1424	5
1241	5
1980	4
1702	4
1521	4
1904	4
993	4
2560	4

1072	4
662	4
1046	4
1711	4
946	4
1842	4
1837	4
1449	4
1466	4
1067	4
1037	4
2160	4
460	4
1259	4
418	4
2070	4
1864	4
1238	4
1111	4
1965	4
921	4
1091	4
1394	4
955	4
755	4
972	4
1367	4
3520	4
1494	4
1202	4
1929	4
3850	4
1537	4
1708	4
1752.12	4
2830	4
8000	4
1169	4
2640	4
966	4
1695	4
2805	4
1795	4
1343	4
1219	4
992	4
3700	4
1353	4
4750	4
1938	4
1081	4
1034	4
1171	4
814	4
1201	4
1258	4

1158	4
1291	4
877	4
1059	4
2450	4
360	4
927	4
1698	4
914	4
1616	4
1976	4
1386	4
1608	4
2439	4
1891	4
996	4
1061	4
1885	4
3436	4
1732	4
1611	4
1236	4
2240	4
1332	4
1945	4
1008	4
12000	4
2135	4
967	4
1933	4
1073	4
1767	4
497	4
595	4
1118	4
1318	4
902	4
1975	4
602	4
795	4
861	4
1563	4
10000	4
1234	4
1776	4
1203	4
1134	4
1074	4
1407	4
1432	4
665	4
1051	4
416	4
4100	4
1504	4
1517	4

1592	4
1138	4
1634	4
2273	4
1058	4
1596	4
1279	4
1489	4
2760	4
1806	4
1334	4
775	4
580	4
884	4
1104	4
1207	4
1881	4
1079	4
1459	4
1149	4
1569	4
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963	1
769	1
1712	1
1918	1
2113	1
2386	1
1210 - 1477	1
892	1
4355	1
167Sq. Meter	1
908	1
3369 - 3464	1
1574	1
1125 - 1500	1
813	1
3884	1
3876	1
897	1
2776	1
3480	1
1	1
2372	1
2775	1
706	1
4356	1
2792	1
3080	1
787	1
660 - 700	1
6613	1
502	1

```
3339                    1
1483                    1
2162                    1
1733                    1
1974                    1
385 - 440               1
2631                    1
2448                    1
4830                    1
770 - 841               1
2171                    1
42000                   1
1793                    1
1551                    1
3563                    1
462                     1
3Cents                  1
36000                   1
1113.12                 1
621                     1
981                     1
2383                    1
555                     1
2138                    1
9000                    1
910.2                   1
4960                    1
3179                    1
2928                    1
1932                    1
2182                    1
2416                    1
3905                    1
2107                    1
2431                    1
4640                    1
818                     1
1288                    1
2155                    1
300                     1
1778                    1
2006.8                  1
1922                    1
888 - 1290              1
2370                    1
5                       1
1004                    1
5985                    1
3580                    1
2461                    1
1437                    1
4689                    1
Name: count, dtype: int64
```

In [18]:

```
# Xử lí các giá trị của trường total_sqft và lưu vào một biến tạm
total_sqft_float = []
```

```
for str_val in df3['total_sqft']:
  try:
    total_sqft_float.append(float(str_val))
  except:
    try:
      temp = []
      temp = str_val.split('-')
      total_sqft_float.append((float(temp[0])+float(temp[-1]))/2)
    except:
      total_sqft_float.append(np.nan)
df4 = df3.reset_index(drop=True)
```

In [19]:

```
# Thêm trường total_sqft_float
df5 = df4.join(pd.DataFrame({'total_sqft_float':total_sqft_float}))
df5.head() #Quan sát kết quả sau khi xử lí
```

Out[19]:

	area_type	availability	location	size	total_sqft	bath	balcony	price	tc
0	Super built-up Area	19-Dec	Electronic City Phase II	2 BHK	1056	2.0	1.0	39.07	
1	Plot Area	Ready To Move	Chikka Tirupathi	4 Bedroom	2600	5.0	3.0	120.00	
2	Built-up Area	Ready To Move	Uttarahalli	3 BHK	1440	2.0	3.0	62.00	
3	Super built-up Area	Ready To Move	Lingadheeranahalli	3 BHK	1521	3.0	1.0	95.00	
4	Super built-up Area	Ready To Move	Kothanur	2 BHK	1200	2.0	1.0	51.00	

In [20]:

```
# Xóa cột "total_sqft"
df5 = df5.drop('total_sqft', axis='columns')
df5.shape
```

Out[20]:

```
(13246, 8)
```

In [21]:

```
# Xem xét lại thông tin về các giá trị null
df5.isnull().sum()
```

Out[21]:

```
area_type           0
availability        0
location            0
size                0
bath                0
balcony             0
price               0
total_sqft_float   46
dtype: int64
```

In [22]:
```
# Loại bỏ các hàng có giá trị null
df6 = df5.dropna()
df6.shape
```

Out[22]:
```
(13200, 8)
```

In [23]:
```
# Xem lại thông tin của dataframe
df6.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 13200 entries, 0 to 13245
Data columns (total 8 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   area_type         13200 non-null  object
 1   availability      13200 non-null  object
 2   location          13200 non-null  object
 3   size              13200 non-null  object
 4   bath              13200 non-null  float64
 5   balcony           13200 non-null  float64
 6   price             13200 non-null  float64
 7   total_sqft_float  13200 non-null  float64
dtypes: float64(4), object(4)
memory usage: 928.1+ KB
```

2. Làm việc với feature: *size*

In [24]:
```
# Quan sát sự phân bố giá trị của trường 'size' với value_counts
df6['size'].value_counts()
```

Out[24]:
```
size
2 BHK         5192
3 BHK         4277
4 Bedroom      816
4 BHK          574
3 Bedroom      541
1 BHK          527
2 Bedroom      325
5 Bedroom      293
6 Bedroom      190
1 Bedroom      100
7 Bedroom       83
8 Bedroom       83
5 BHK           56
9 Bedroom       45
6 BHK           30
7 BHK           17
1 RK            13
10 Bedroom      12
9 BHK            7
8 BHK            5
11 BHK           2
11 Bedroom       2
10 BHK           2
14 BHK           1
13 BHK           1
12 Bedroom       1
27 BHK           1
43 Bedroom       1
16 BHK           1
19 BHK           1
18 Bedroom       1
Name: count, dtype: int64
```

In [25]:
```
# Chuyển thuộc tính số phòng từ dạng category về dạng numeric
size_int = []
for str_val in df6['size']:
  temp=[]
  temp = str_val.split(" ")
  try:
    size_int.append(int(temp[0]))
  except:
    size_int.append(np.nan)
    print("Noise = ", str_val)
```

In [26]:
```
# Đánh lại index cho các hàng
df6 = df6.reset_index(drop=True)
```

In [27]:
```
# Thêm trường dữ liệu số phòng (bhk)
df7 = df6.join(pd.DataFrame({'bhk':size_int}))
df7.shape
```

Out[27]:
```
(13200, 9)
```

In [28]:
```
# In ra kết quả thực hiện các thao tác kể trên?
df7.tail()
```

Out[28]:

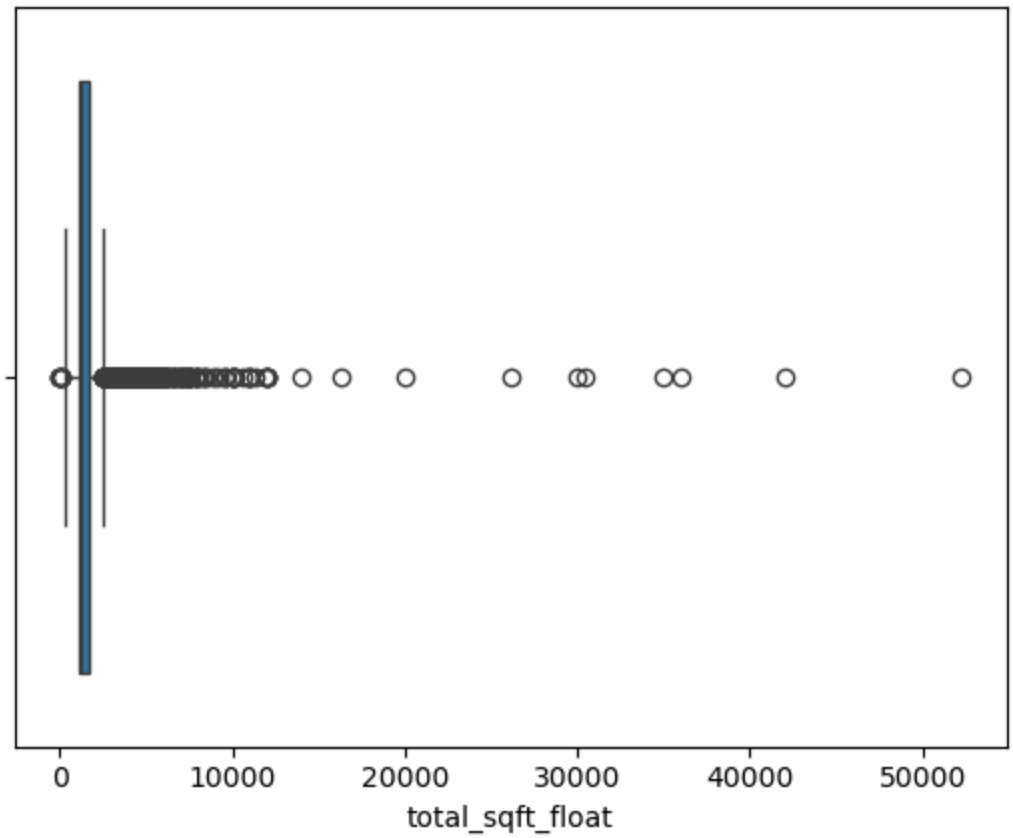
	area_type	availability	location	size	bath	balcony	price	total_sqf
13195	Built-up Area	Ready To Move	Whitefield	5 Bedroom	4.0	0.000000	231.0	
13196	Super built-up Area	Ready To Move	Richards Town	4 BHK	5.0	1.584376	400.0	
13197	Built-up Area	Ready To Move	Raja Rajeshwari Nagar	2 BHK	2.0	1.000000	60.0	
13198	Super built-up Area	18-Jun	Padmanabhanagar	4 BHK	4.0	1.000000	488.0	
13199	Super built-up Area	Ready To Move	Doddathoguru	1 BHK	1.0	1.000000	17.0	

4.1.3. Phát hiện ngoại lệ (outlier) và loại bỏ

- Dựa trên biểu đồ boxplot/ hoặc công cụ khác để phát hiện và loại bỏ các điểm ngoại lai:

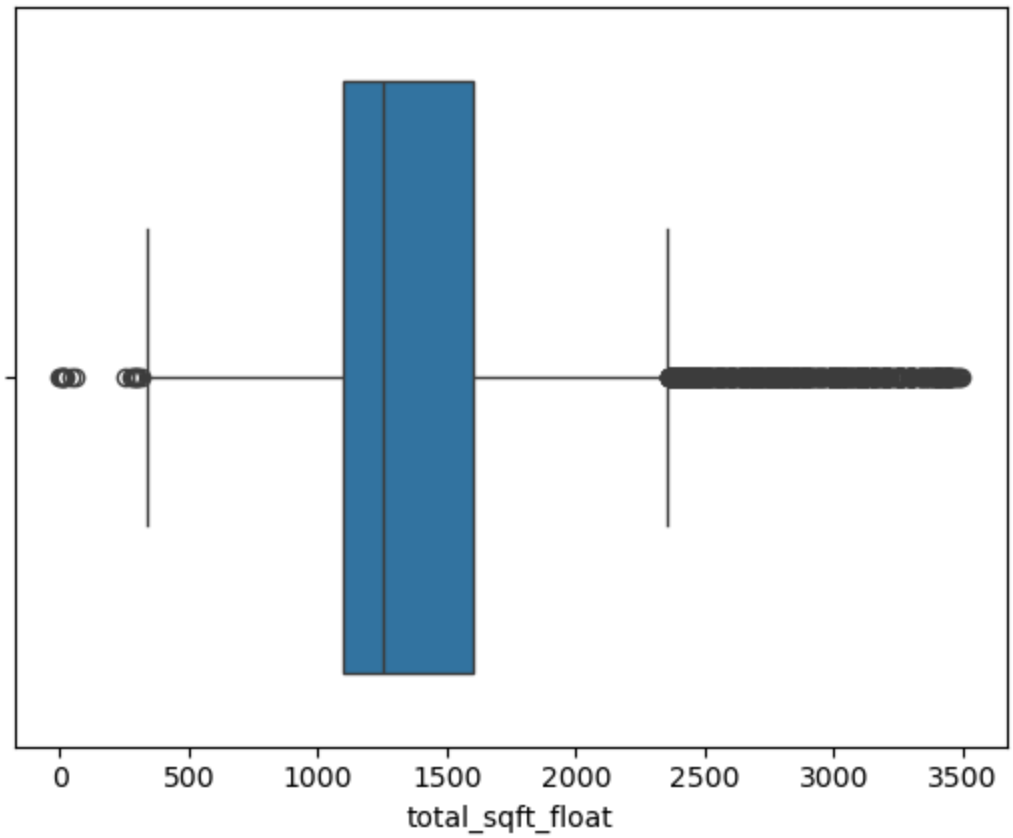
In [29]:
```
# Xem xét trường diện tích:
sns.boxplot(x = df7['total_sqft_float'])
```

Out[29]:
```
<Axes: xlabel='total_sqft_float'>
```



In [30]:
```
# Chọn ngưỡng diện tích là 3500 để xem xét
df_temp7 = df7[df7['total_sqft_float'] < 3500]
sns.boxplot(x = df_temp7['total_sqft_float'])
```

Out[30]: <Axes: xlabel='total_sqft_float'>

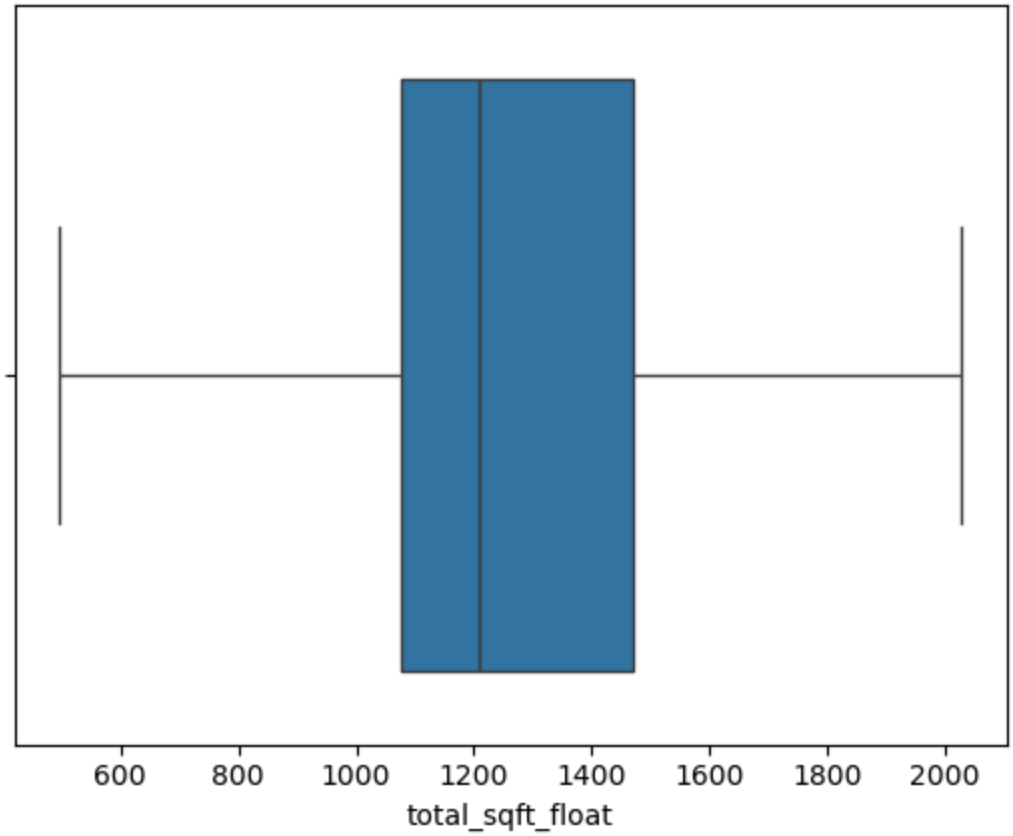


In [32]:
```
Q1 = df7['total_sqft_float'].quantile(0.25)
Q3 = df7['total_sqft_float'].quantile(0.75)
IQR = Q3 - Q1
low_threshold = Q1 - 1.5*IQR
up_threshold = Q3 + 1.5*IQR
print(low_threshold, up_threshold)
```

```
242.0 2530.0
```

In [38]:
```
# Loại bỏ đi các điểm dữ liệu có diện tích nhiễu
# (Q1 - 1.5 * IQR or Q3 + 1.5 * IQR)
df8 = df7[(df7['total_sqft_float'] > low_threshold+250) & (df7['total_sqft_float']
sns.boxplot(x = df8['total_sqft_float'])
df8.shape
```

Out[38]: (11128, 9)



In [39]:
```
# Tạo thêm trường dữ liệu price_per_sqft (giá/ diện tích feet vuông)
df8 = df8.reset_index(drop=True)
df8['price_per_sqft'] = df8['price']*100000 / df8['total_sqft_float']
df8.head()
```

Out[39]:

	area_type	availability	location	size	bath	balcony	price	total_sqft_float	bl
0	Super built-up Area	19-Dec	Electronic City Phase II	2 BHK	2.0	1.0	39.07	1056.0	
1	Built-up Area	Ready To Move	Uttarahalli	3 BHK	2.0	3.0	62.00	1440.0	
2	Super built-up Area	Ready To Move	Lingadheeranahalli	3 BHK	3.0	1.0	95.00	1521.0	
3	Super built-up Area	Ready To Move	Kothanur	2 BHK	2.0	1.0	51.00	1200.0	
4	Super built-up Area	Ready To Move	Whitefield	2 BHK	2.0	1.0	38.00	1170.0	

In [40]:
```
df8.price_per_sqft.describe()
```

Out[40]:
```
count    11128.000000
mean      6193.020133
std       3703.924767
min       1166.666667
25%       4166.666667
50%       5183.169954
75%       6667.625072
max      76923.076923
Name: price_per_sqft, dtype: float64
```

4.1.4. Chuyển dữ liệu từ dạng category sang numeric cho các thuộc tính còn lại

In [41]:
```
df8.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11128 entries, 0 to 11127
Data columns (total 10 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   area_type         11128 non-null  object
 1   availability      11128 non-null  object
 2   location          11128 non-null  object
 3   size              11128 non-null  object
 4   bath              11128 non-null  float64
 5   balcony           11128 non-null  float64
 6   price             11128 non-null  float64
 7   total_sqft_float  11128 non-null  float64
 8   bhk               11128 non-null  int64
 9   price_per_sqft    11128 non-null  float64
dtypes: float64(5), int64(1), object(4)
memory usage: 869.5+ KB
```

In [43]:
```
value_count(df8.iloc[:,:4])
```

```
area_type
Super built-up  Area    7736
Built-up  Area          1888
Plot  Area              1434
Carpet  Area              70
Name: count, dtype: int64
--------------------------------
availability
Ready To Move    8819
18-May            252
18-Dec            245
18-Apr            243
18-Aug            187
19-Dec            150
18-Jul            124
18-Mar             96
18-Jun             88
20-Dec             84
21-Dec             79
19-Mar             76
18-Feb             51
18-Nov             42
18-Sep             39
18-Jan             38
19-Jul             35
18-Oct             32
19-Jan             29
19-Jun             27
20-Jan             24
17-Dec             23
17-Oct             21
22-Jun             19
19-Apr             16
21-Jun             15
19-Sep             15
21-Mar             14
19-Aug             13
21-Jan             12
20-Aug             12
17-Jun             12
22-Dec             11
17-Jul             11
21-May             11
20-Jun             11
17-Sep             10
19-Oct              9
17-Nov              9
21-Jul              8
17-Aug              8
19-Nov              7
20-Oct              7
21-Sep              6
22-Jan              6
21-Oct              6
17-May              6
21-Aug              5
```

```
17-Apr              5
19-May              5
20-Nov              4
22-May              4
21-Feb              4
17-Mar              4
20-Sep              4
22-Mar              3
19-Feb              3
21-Nov              3
16-Dec              2
20-Mar              2
15-Nov              2
20-Apr              2
22-Nov              2
17-Feb              1
16-Jan              1
16-Nov              1
15-Aug              1
16-Oct              1
15-Dec              1
16-Sep              1
20-Feb              1
15-Oct              1
20-Jul              1
20-May              1
15-Jun              1
16-Jul              1
14-Nov              1
16-Mar              1
14-Jul              1
Name: count, dtype: int64
--------------------------------
location
Whitefield                                            411
Sarjapur  Road                                        311
Electronic City                                       294
Kanakpura Road                                        248
Thanisandra                                           208
Yelahanka                                             184
Uttarahalli                                           182
Raja Rajeshwari Nagar                                 164
Marathahalli                                          148
Bannerghatta Road                                     138
7th Phase JP Nagar                                    132
Haralur Road                                          130
Electronic City Phase II                              127
Hennur Road                                           126
Hebbal                                                118
Chandapura                                             94
Bellandur                                              87
Electronics City Phase 1                               85
Hoodi                                                  80
Begur Road                                             79
KR Puram                                               78
Yeshwanthpur                                           73
```

Harlur	69
Kasavanhalli	68
Hormavu	68
Hosa Road	68
Banashankari	66
Varthur	65
Ramamurthy Nagar	64
Kothanur	60
Rajaji Nagar	60
Sarjapur	59
Nagarbhavi	59
Kengeri	58
Kaggadasapura	58
Jakkur	57
TC Palaya	57
Akshaya Nagar	57
Koramangala	56
Budigere	54
8th Phase JP Nagar	52
JP Nagar	52
Panathur	51
Bisuvanahalli	51
Rachenahalli	49
Hennur	49
Hulimavu	47
Jigani	47
Mysore Road	46
HSR Layout	46
Old Madras Road	46
Gottigere	46
Jalahalli	46
Balagere	45
CV Raman Nagar	43
Vittasandra	42
Subramanyapura	42
Horamavu Agara	41
Ramagondanahalli	41
Hosur Road	40
9th Phase JP Nagar	40
Kundalahalli	40
Kanakapura	40
Brookefield	38
Vidyaranyapura	38
Bhoganhalli	38
Bommasandra	37
Kudlu Gate	37
Kengeri Satellite Town	37
Vijayanagar	37
Hegde Nagar	36
Kadugodi	36
5th Phase JP Nagar	35
Green Glen Layout	35
Channasandra	35
Sahakara Nagar	35
Lakshminarayana Pura	35
Kalena Agrahara	33

Devanahalli	32
R.T. Nagar	32
Tumkur Road	32
Bommanahalli	31
Anekal	31
Ananth Nagar	30
Doddathoguru	30
Chikkalasandra	30
Yelahanka New Town	29
Mahadevpura	29
Kammasandra	29
Kudlu	28
Horamavu Banaswadi	27
Anandapura	27
Thigalarapalya	26
Indira Nagar	26
Kothannur	26
Dodda Nekkundi	26
Somasundara Palya	25
Babusapalaya	25
Talaghattapura	25
Bommasandra Industrial Area	25
Margondanahalli	25
Kambipura	24
Singasandra	24
Attibele	24
Malleshwaram	23
Choodasandra	23
Ambalipura	23
Banashankari Stage III	23
BTM 2nd Stage	23
Munnekollal	22
Kathriguppe	22
Sonnenahalli	22
Basavangudi	22
Kogilu	22
Kodichikkanahalli	22
Magadi Road	22
Padmanabhanagar	21
Hoskote	21
Abbigere	21
6th Phase JP Nagar	21
Ambedkar Nagar	21
Battarahalli	21
Gubbalala	21
Kumaraswami Layout	21
Hebbal Kempapura	21
Amruthahalli	21
Lingadheeranahalli	20
2nd Stage Nagarbhavi	20
Rayasandra	20
Gunjur	20
Poorna Pragna Layout	19
Thubarahalli	19
Yelachenahalli	19
Hosakerehalli	19

Pai Layout	18
HRBR Layout	18
Kaval Byrasandra	18
HBR Layout	18
Domlur	17
Seegehalli	17
Sanjay nagar	17
Garudachar Palya	17
Frazer Town	17
Malleshpalya	16
Sector 2 HSR Layout	16
Kereguddadahalli	16
Kaggalipura	16
Billekahalli	16
Parappana Agrahara	16
Basaveshwara Nagar	16
Anjanapura	16
Mallasandra	16
Binny Pete	16
1st Phase JP Nagar	16
Kaikondrahalli	16
Dasanapura	15
Kenchenahalli	15
OMBR Layout	15
Kasturi Nagar	15
Nagavarapalya	15
Kannamangala	15
Kammanahalli	15
NGR Layout	14
Banaswadi	14
Dasarahalli	14
Ardendale	14
Sarjapura - Attibele Road	14
Kalyan nagar	14
Devarachikkanahalli	14
Varthur Road	14
Jalahalli East	14
Begur	14
EPIP Zone	13
Bannerghatta	13
Yelenahalli	13
Gollarapalya Hosahalli	13
Chamrajpet	13
Sultan Palaya	12
GM Palaya	12
Murugeshpalya	12
BTM Layout	12
Kodigehaali	12
Banashankari Stage II	12
Dommasandra	12
Banashankari Stage V	12
AECS Layout	12
Cox Town	12
Doddakallasandra	12
Old Airport Road	12
Bharathi Nagar	12

Ulsoor	12
Badavala Nagar	12
Banashankari Stage VI	12
2nd Phase Judicial Layout	11
Judicial Layout	11
Kadubeesanahalli	11
Laggere	11
Arekere	11
ISRO Layout	11
BEML Layout	11
Vasanthapura	11
Sompura	11
Vishwapriya Layout	11
Marsur	11
Kodigehalli	11
Pattandur Agrahara	11
Nagappa Reddy Layout	10
Tindlu	10
Sector 7 HSR Layout	10
Ganga Nagar	10
Basapura	10
1st Block Jayanagar	10
Mico Layout	10
ITPL	10
Kalkere	10
Shivaji Nagar	10
Mahalakshmi Layout	10
NRI Layout	10
Gunjur Palya	10
LB Shastri Nagar	10
Chikkabanavar	10
Shampura	10
Banjara Layout	9
Prithvi Layout	9
Bommenahalli	9
Vignana Nagar	9
Karuna Nagar	9
Banagiri Nagar	9
Vishveshwarya Layout	9
Neeladri Nagar	9
B Narayanapura	9
Chennammana Kere	9
5th Block Hbr Layout	9
Peenya	9
Iblur Village	9
KUDLU MAIN ROAD	9
Chandra Layout	9
Thyagaraja Nagar	9
Gollahalli	9
Nagadevanahalli	9
Mathikere	9
Naganathapura	9
Cooke Town	8
Ejipura	8
Hongasandra	8
Nehru Nagar	8

Kaverappa Layout	8
Basavanapura	8
Vajarahalli	8
BTM 1st Stage	8
Sidedahalli	8
Reliaable Tranquil Layout	8
Nagasandra	8
Volagerekallahalli	8
Medahalli	8
Nagavara	8
Hoysalanagar	8
Kamakshipalya	8
Nallurhalli	8
Narayanapura	8
Seetharampalya	8
Dodda Nekkundi Extension	8
Roopena Agrahara	8
Giri Nagar	8
Benson Town	8
Ittamadu	8
Huskur	8
Shikaripalya	8
Konanakunte	7
AGS Layout	7
Gopalkrishna Nagar	7
Jeevan bima nagar	7
Ambedkar Colony	7
Kodbisanhalli	7
Doctors Layout	7
Hessarghatta	7
Byrasandra	7
Sector 1 HSR Layout	7
Keshava Nagar	7
2nd Phase JP Nagar	7
Jalahalli West	7
3rd Block Hrbr Layout	7
Lingarajapuram	7
Canara Bank Colony	7
Green View Layout	7
Hennur Gardens	7
Rajiv Nagar	7
Akshaya Vana	7
Akshayanagara East	7
Thirumenahalli	7
Ashwath Nagar	7
HOSUR MAIN ROAD	7
Tejaswini Nagar	7
Kanaka Nagar	6
Wilson Garden	6
Madiwala	6
8th block Koramangala	6
Byatarayanapura	6
Doddanakundi Industrial Area 2	6
Cambridge Layout	6
HAL 2nd Stage	6
Dodsworth Layout	6

Doddakammanahalli	6
Kacharakanahalli	6
Devasthanagalu	6
Mahalakshmi Puram	6
Austin Town	6
2nd Block Hrbr Layout	6
Manayata Tech Park	6
Devarabeesana Halli	6
Abbaiah Reddy Layout	6
Sarjapur Road,	6
Hoodi Circle,	6
Byrathi Village	6
Rustam Bagh Layout	6
Maragondanahalli	6
Jakkur Plantation	6
Basaveshwara Nagar Yelahanka	6
Nelamangala	6
Varanasi	6
Rajiv Gandhi Nagar	6
Suraksha Nagar	6
Doddakannelli	6
Motappa Layout	6
Singapura Village	5
Bhuvaneshwari Nagar	5
New Thippasandra	5
Sarakki Nagar	5
Doddabanahalli	5
Pattanagere	5
Hennur Bande	5
KR Garden	5
9th Block Jayanagar	5
Kadabagere	5
Nayandanahalli	5
Coconut Grove Layout	5
Kallumantapa	5
Dodda Banaswadi	5
Kadugondanahalli	5
Devi Nagar	5
Govindpura	5
Richards Town	5
Mallathahalli	5
Outer Ring Road East	5
Byagadadhenahalli	5
Victoria Layout	5
Cholanayakanahalli	5
Cottonpet	5
1st Block Koramangala	5
Chikkadunnasandra	5
1st Block HRBR Layout	5
Nagaraja Garden	5
Sunkadakatte	5
Maithri Layout	5
Channasandra Layout	5
Bagalakunte	5
A Narayanapura	5
Amruthnagar	5

ITI Layout	5
Jakkuru Layout	5
Samethanahalli	5
Chelekare	5
Richmond Town	5
Vishwanatha Nagenahalli	5
RMV 2nd Stage	5
Maruthi Sevanagar	5
Silk Board	5
Srirampura	5
RMV Extension	5
Ashirvad Colony	5
Shree Ananth Nagar Layout	5
Anand Nagar	5
Vasantha Vallabha Nagar	5
Kundalahalli Colony	5
Nagondanahalli	5
Bank Of Baroda Colony	5
MS Pallya	5
Srinivasa Nagar	5
NS Palya	4
Hosahalli Extension	4
Doopanahalli	4
Yemlur	4
Hiremath Layout	4
Suddaguntepalya	4
Jakkasandra Extension	4
Udaya Nagar	4
Udayapur Village	4
Doddabommasandra	4
Chikkasandra	4
Herohalli	4
Chinnapanahalli	4
Ramanjaneyanagar	4
Doddagubbi	4
Maruthi Nagar	4
Kada Agrahara	4
Kattigenahalli	4
Kodihalli	4
BSM Extension	4
tc.palya	4
JCR Layout	4
Dooravani Nagar	4
Doddanekundi	4
Alfa Garden Layout	4
Lakshmi Layout	4
Glass Factory Layout	4
Brooke Bond First Cross	4
Tala Cauvery Layout	4
Off Sarjapur Road,	4
Bagalur	4
Geddalahalli	4
Tata Nagar	4
Bellari Road	4
Mylasandra	4
Vijaya Bank Layout	4

Virat Nagar	4
Kumarapalli	4
Raja Rajeshwari Nagar 5th Stage	4
Kurubarahalli	4
Chikkathoguru	4
MCECHS layout	4
Pragathi Nagar	4
Amblipura	4
Silver Springs Layout	4
Akshya Nagar	4
Kodathi	4
Jayanagar	4
Basavanna Nagar	4
Lake City	4
Nagawara Junction	4
T Dasarahalli	4
Upkar Layout	4
Ashwathnagar	4
Mysore Highway	4
Immadihalli	4
Nyanappana Halli	4
Vinayaka Nagar	4
HMT Layout	4
AMS Layout	4
RWF West Colony	4
Sai Gardens	4
Hanumanth Nagar	4
Chennappa Layout	4
VGP Layout	4
Judicial Layout, Kanakapura Road,	4
Divya Unnathi Layout	4
Manjunath Nagar	4
Stage-4 Bommanahalli	4
Lavakusha Nagar	4
Nagashetty Halli	4
Puttanahalli	4
Ramamurthy Nagar Extension	4
New Gurappana Palya	4
Belathur	4
Shanti Nagar	4
Agrahara Dasarahalli	3
Manorayana Palya	3
Kachanayakanahalli	3
Shantiniketan Layout	3
Kenchenhalli	3
Gopalapura	3
4th Block Jayanagar	3
4th Phase JP Nagar	3
Gowdanapalya	3
Sathya Layout	3
J C Nagar	3
Kaveri Nagar	3
Srirampuram	3
Doddaballapur	3
Gokula Extension	3
Gkvk Layout	3

2nd Block Jayanagar	3
Sanne Amanikere	3
Singanayakanahalli	3
Kamala Nagar	3
Jnana Ganga Nagar	3
Nobo Nagar	3
6th block Koramangala	3
Jayanti Nagar	3
Vishwapriya Nagar	3
Sampangi Rama Nagar	3
Esther Enclave Layout	3
BCC Layout	3
Adityanagar	3
Sarvabhouma Nagar	3
SMV layout	3
Lavelle Road	3
Dinnur	3
Vasanth nagar	3
Dollars Colony	3
Telecom Layout	3
Bhagyalakshmi Avenue	3
1st Stage Indira Nagar	3
Kengeri Hobli	3
OLd Gurappanapalya	3
Rajarajeshwari nagar	3
Jinkethimmanahalli	3
Kuvempu Nagar	3
Hagadur	3
Nandini Layout	3
Thippasandra	3
B Channasandra	3
Ayappa Nagar	3
Arehalli	3
Kithaganur	3
Lakshmipura	3
Pampa Extension	3
Basavanagara	3
Kothnur Narayanapura	3
Venugopal Reddy Layout	3
T.C PALYA	3
Mangammanapalya	3
Seshadripuram	3
KSRTC Layout	3
Tunganagara	3
BEL Road	3
3rd Block Banashankari	3
Kumbena Agrahara	3
Carmelaram	3
Kammagondahalli	3
Venkatadri Layout	3
Ashwini layout	3
Thurahalli	3
Vittal Nagar	3
Moodalapalya	3
Raghuvanahalli	3
RPC layout	3

Sadaramangala	3
Gattahalli	3
Singena Agrahara	3
Konanakunte Cross	3
Kumara Park	3
Kavika Layout	3
Tirumanahalli	3
Canara Bank Layout	3
VHBCS Layout	3
Chikka Tirupathi	3
Hegganahalli	3
Basava Nagar	3
Shanthala Nagar	3
Dasappa Layout	3
Srinagar	3
Govindapura	3
Ankappa Layout	3
Poornapragna Housing Society Layout	3
Brindavan Nagar	3
Tavarekere	3
Vinayak Nagar	3
Shankarapuram	3
Daadys Gaarden Layout	3
Anjappa Layout	3
Crimson Layout	3
BTM 4th Stage	3
Kirloskar Layout	3
5th Stage BEML Layout	3
Virupakshapura	3
Baba Nagar	3
Defence Colony	3
Rajarajeshwari Nagara	3
Narayana Nagar 1st Block	3
Siddapura	3
Laxmi Sagar Layout	3
Nanjappa Layout	3
P&T Layout	3
Surabhi Layout	3
Dairy Circle	3
Shettigere	3
Atmananda Colony	3
8th Block Jayanagar	3
Tasker Town	3
Devarachikkanahalli	3
Belatur	3
Palace Guttahalli	3
Ilyas Nagar	3
Chikka Banaswadi	3
Gulimangala	3
Bethel Nagar	3
Anantapura	3
Himagiri Meadows	3
Kashi Nagar	3
4th Block Koramangala	3
Veerannapalya	3
Tigalarpalya	2

Munivenkatppa Layout	2
Thanisandra	2
Celebrity Paradise Layout	2
Jai Bheema Nagar	2
2nd Block Bel Layout	2
Venkatapura	2
Hadosiddapura	2
M V Extenstion	2
Anand nagar	2
Langford Town	2
Anjana Nagar	2
Akshayanagara West	2
Veersandra	2
Upadhyaya Layout	2
Marenahalli	2
MLA Layout	2
Janatha Colony	2
Kyalasanahalli	2
Ramanashree Enclave	2
Harsha Layout	2
Gidada Konnenahalli	2
cooketown	2
Subash Nagar	2
Shingapura	2
Thomas Town	2
Indra Nagar	2
7th Block Jayanagar	2
Challaghatta	2
Varsova Layout	2
Lottegolla Halli	2
Raghavendra Layout	2
Konena Agrahara	2
Weavers Colony	2
Kodipalya	2
Krishna Nagar	2
BCMC Layout	2
Gattigere	2
Ramakrishnappa Layout	2
1st Block BEL Layout	2
Anugrah Layout	2
Makali	2
Uday Nagar	2
Devara Jeevanahalli	2
Attur Layout	2
Marasandra	2
Naidu Layout	2
Ullal Uppanagar	2
Electronic city Phase 1,	2
Bikasipura	2
Vimanapura	2
Guni Agrahara	2
Kempegowda Nagar	2
Whitefield,	2
Yarandahalli	2
Gandhi Bazar	2
Neelamangala	2

Garebhavipalya	2
Ramesh Nagar	2
Vasanthpura	2
Guddadahalli	2
Kothnoor Dinne	2
Rajarajeshwarinagar	2
Madavara	2
Avalahalli	2
Dena Bank Colony	2
5th Block Jayanagar	2
GD Layout	2
Rajasree Layout	2
JP Nagar 7th Phase,	2
Kariyammana Agrahara	2
Bidrahalli	2
KEB Colony	2
Sri Sai Layout	2
Jayamahal	2
2nd Stage Arekere Mico Layout	2
Coffee Board Layout	2
Chikkakannalli	2
D Group Employees Layout	2
Phase 1 Kammasandra	2
Syndicate Bank Colony	2
Sri Venkateshpura Layout	2
Sri Balaji Krupa Layout	2
Prashanth Nagar	2
Gaurava Nagar	2
Muneshwara Nagar	2
SRINIVASAPURA	2
Pulkeshi Nagar	2
Green Garden Layout	2
Sunder Ram Shetty Nagar	2
BHEL Layout	2
Doddabidrakallu	2
Bidadi	2
Ullal Road	2
Varthur Road,	2
Postal Colony	2
Devasandra Extension	2
Devanahalli Int. Airport	2
Vyalikaval	2
Geleyara Balaga Layout	2
Balaji Gardens Layout	2
Vadarpalya	2
ECC Road, Whitefield,	2
Vibhutipura Extension	2
Sampigehalli	2
Shanthi Layout	2
Sneha Colony	2
Omkar Nagar	2
SBM Colony	2
Patelappa Layout	2
Vijinapura	2
Remco Bhel Layout	2
Panduranga Nagar	2

Mahaganapathy Nagar	2
Ngef Layout	2
Amam Enclave Layout	2
Hosapalya	2
HAL 3rd Stage	2
Annaiah Reddy Layout	2
Hoodi Layout	2
Manjunatha Layout	2
3rd Block Jayanagar	2
Jnanabharathi Layout	2
Bagalur Main Road	2
Green Woods Layout	2
Someshwara Layout	2
Devanahalli Road	2
Coconut Garden	2
Friends Colony	2
Mathikere Extension	2
Shirdi Sai Layout	2
Dr Shivarama Karantha Nagar	2
Dodda Kempaiah Layout	2
Jyothi Nagar	2
Mariyannapalya	2
Chennammanakere Achukattu	2
Kullappa Colony	2
Hennagara	2
Anwar Layout	2
Palace Road	2
Chokkanahalli	2
Hallehalli	2
Amarjyothi Colony	2
Classic Paradise Layout	2
Bhuvaneswari Nagar	2
Cleveland Town	2
manyata park	2
KR Layout	2
Soundarya Layout	2
Raghavendra Nagar	2
Domlur Layout	2
Sarvobhogam Nagar	1
basaveshwarnagar	1
Banashankari 3rd stage, Vivekanandanagar	1
Behind Don Bosco Church	1
Pattegarhpalya	1
Rahmath Nagar	1
Near International Airport	1
manyata	1
Sugama Layout	1
Raja Rajashweri Nagar	1
Banaswadi	1
PNS Layout	1
kadubisnahalli	1
bsk 6th stage 2ad block near sri conversation hall	1
SK Garden	1
1st Block HBR Layout	1
HSR Layout 7th sector,	1
Neelasandra	1

Xavier Layout	1
Air View Colony	1
Yediyur	1
Somanna Garden	1
Park View Layout	1
Havanur extension	1
Bettadasanapura	1
Halanayakanahalli	1
Chikkanahalli	1
Prasanna layout Herohalli	1
Annapurneshwari Nagar	1
poornaprajna layout	1
Arishinakunte	1
akshaya nagar t c palya	1
12th cross srinivas nagar banshankari 3rd stage	1
Bandepalya	1
mvj engineering college	1
Beml layout, Rajarajeshwari nagar	1
LIC Colony	1
Annapoorneshwari Layout, JP nagar 7th phase	1
Bilal Nagar	1
Veerabhadra Nagar	1
Shetty Halli	1
Vidhyanagar Cross	1
Punappa Layout	1
Chellikere	1
Electronic City Phase 1,	1
Jakkasandra	1
Navodaya Nagar	1
T C Palya main Road	1
Chuchangatta Colony	1
Kengeri Satellite Town KHB Apartment	1
Subhash Nagar	1
Jogupalya	1
Hommadevanahalli	1
Gulakamale	1
Nagarbhavi Garden Villas Layout	1
Indira Nagar Stage 2	1
T R Mill Road	1
Chikbasavanapura	1
Zuzuvadi	1
Kanakapura Main Road	1
Basavangudi	1
Suragajakkanahalli	1
Mallappa Layout	1
Duddanahalli	1
elachenahalli	1
Kanakapur main road	1
Byrasandra Extension	1
Masjid e Alkareem	1
3rd Stage Raja Rajeshwari Nagar	1
Dominic Layout	1
Suryanagar	1
KAMAKIYA	1
Chaitanya Ananya	1
Kanakapura main Road	1

Wheelers Road	1
Kengeri Satellite Town Stage II	1
Devarabeesana Halli	1
Banashankari 6th stage , 2nd block	1
Duvasapalya	1
Mudalpalaya	1
Vaderahalli	1
Addischetan Layout	1
PC Palaya	1
7th Block Koramangala	1
kamanahalli main road	1
Virudhu Nagar	1
4th Block HBR Layout	1
Michael Palaya	1
RBI Layout	1
Aavalahalli	1
Akash Nagar	1
Shauhardha Layout	1
Soppahalli	1
Harohalli	1
Andrahalli	1
Bharat Nagar	1
Electronic city phase 1,	1
Anathanagar	1
Ramchandrapuram	1
Kathriguppe IV Phase	1
vinayakanagar	1
Jaraganahalli Jp Nagar Post	1
adigondanhalli	1
Off Bannergatta Road	1
Chambenahalli	1
Kanakapura Rod	1
MEI layout, Bagalgunte	1
Prakash Nagar	1
AECS LAYOUT A-BLOCK Singasandra	1
Jeevanhalli	1
BAGUR ROAD	1
Thirupalya	1
Hoskote near	1
Adugodi	1
Raja Rajeshwari Nagara	1
Yeshwanthpur Industrial Suburb	1
Rajapura	1
RK Colony	1
Rammana Layout	1
Nanjappa Layout Vidyaranyapura	1
kg halli jalhalli west	1
Vasatha Vallbha Nagar	1
Kempapura	1
Pillanna Gardens	1
Balepet	1
Electronic City	1
Banashankari 2 nd Stage	1
Venkateswara Nagar	1
Gollarahatti	1
RMC YARD	1

Panathur Road,	1
Vivek Nagar	1
Shanthi Pura	1
Ashok Nagar	1
Pillahalli	1
Banashankari 6th Stage	1
Sabari Nagar	1
Chamundi Nagar	1
Aishwarya Crystal Layout	1
Hosakerehalli Layout	1
RR Nagar	1
Koppa	1
Queens Road	1
Javarandoddi	1
Vijaypura	1
Bapuji Nagar	1
Satyasaibaba Layout	1
Nanjappa Garden	1
Jay an agar 4 T Block	1
Allalasandra	1
SHANTHINAGAR	1
Jagadish Nagar	1
Ramanagara Channapatna	1
HAL Layout	1
Maheswari Nagar	1
Hsr layout sector3	1
Thyagraj Nagar	1
Bhattarahalli	1
Banagirinagar	1
Medaralli	1
Versova Layout	1
Chikkajala	1
Udayagiri	1
Anantapuram	1
3rd Phase Iti Layout	1
Dwaraka Nagar	1
Brindavan Layout	1
Thanisandra Main Road,	1
Vibuthipura	1
2nd Block Hbr Layout	1
Kathreguppe	1
Rajarajesheari nagar	1
Asthagrama Layout	1
Jayanagar,	1
Whitefield ECC Road	1
Bharathnagar	1
Thirumalashettyhally	1
Maruthi nagar kogilu	1
Billamaranahalli	1
Green Domain Layout	1
Prasanti Nagar	1
Konappana Agrahara	1
HOSUR RMAIN ROAD	1
B Y Raveshwara Nagar	1
Sunkan palya	1
Nandi Durga Road	1

Gayathri Nagar	1
1Kasavanhalli	1
Ananthapura, T C palaya Main Road	1
Kanakapura Road,	1
Hal old airport road	1
Ex-Servicemen Colony Dinnur Main Road R.T.Nagar	1
Sadduguntepalya	1
Narayanappa Garden	1
Howthinarayanappa Garden	1
Viswajit Layout	1
Srigandada Kaval	1
Prestige Sunrise	1
Tippenahalli	1
Thavarekere	1
Sundara Nagar	1
Kudlu Village,	1
Virgonagar	1
Sampige Layout	1
banashankari stage iii sa	1
Kumbalgodu	1
ittamadu	1
Bull Temple Road	1
Keerthi Layout	1
Cambridge road	1
Harappanahalli	1
Bahubali Nagar	1
Rajagopala Nagar	1
RMV	1
Banashankari3rd stage bigbazar	1
Binny Mills Employees Colony	1
JakkurYelahanka	1
Arasanakunte	1
Gaundanapalya	1
Kirloskar layout, Basaveshwarnagar	1
K N Extension	1
BDS Layout	1
Ganga Nagar Extension	1
kanakapura road	1
RK Layout 2nd Stage	1
Goraguntepalya	1
Hanumantha Nagar	1
P Krishnappa Layout	1
Mathikere SBM colony	1
Sahyadri Layout	1
Laxminarayana Layout	1
Kanakadasa Layout	1
Indira Nagar 3rd Stage	1
Kumbhena Agrahara	1
1Hanuman Nagar	1
Narayanappa Layout	1
Lakshminarayanapura, Electronic City Phase 2	1
Sri Kanteshwara Nagar	1
Reliable Woods Layout	1
BEL Layout	1
J P Nagar 7th Phase Ramayya City	1
Kanakapura Road	1

Dhanalakshmi Layout	1
Maragondana Halli, kr puram, old madras road	1
Govindraja Nagar	1
T c palya	1
Chikkabettahalli	1
Dollars Layout	1
Ncpr Industrial Layout	1
Banashankari 6th Stage ,Subramanyapura	1
Popular Colony	1
Sundar Ram Shetty Nagar	1
Sathanur	1
Ganganahalli	1
Defence Layout	1
Brigade Road	1
Kengeri Satellite Town (BDA SITE)	1
Sulthangunta	1
Prakruthi Nagar	1
GB Palya	1
Hennur Busstop	1
Rukmaiah Layout	1
Shivanagar	1
Bettahalsoor	1
V.V Puram	1
Sector 3 HSR Layout	1
Kalasipalya	1
T.C. Palya	1
Maruthi HBCS Layout	1
Bendiganahalli	1
Chikkaballapur	1
Ferrar Nagar	1
Iggalur	1
Williams Town	1
BAGUR	1
M.G Road	1
sapthagiri Layout	1
Alur	1
Bidere Agarahara, Behind Safal market	1
Devarabisanahalli	1
Mango Garden Layout	1
Teachers Colony	1
Craig Park Layout	1
St. John's Road	1
Chowdeshwari Layout	1
Attiguppe	1
Amco Colony	1
Chokkasandra	1
Sree Narayana Nagar	1
Maruthi Layout	1
Kenchanehalli R R Nagar	1
Agara Village	1
NR Colony	1
ITI Employees Layout	1
Bank Avenue	1
3rd Block HBR Layout	1
Kamakya Layout	1
Yemlur, Old Airport Road,	1

Hosahalli	1
Jaya Mahal layout	1
1 Ramamurthy Nagar	1
Subramanyanagar	1
Kadarenahalli	1
K G Colony	1
frazertown	1
T K Reddy Layout	1
Garden Layout	1
Marathalli bridge	1
Malimakanapura	1
Papareddipalya	1
KPC Layout	1
St Thomas Town	1
Ckikkakammana Halli	1
Banashankari 2nd Stage	1
3rd Block Koramangala	1
Hullahalli	1
MRCR Layout	1
Arekempanahalli	1
CR Layout	1
Sonam Layout	1
4th T block Jayanagar	1
Fraser town	1
NTI Layout	1
Nagarbhavi BDA Complex	1
Rajanna Layout	1
Ganesha Block	1
Adarsh Nagar	1
Rainbow Drive	1
Lakkasandra Extension	1
Shakthi Nagar	1
Kalkere Channasandra	1
Lakkasandra	1
Seethappa Layout	1
Bennigana Halli	1
Prakruthi Township	1
ravindra nagar, T.dasarahalli peenya	1
Govindaraja Nagar Ward	1
RTO ullalu	1
Gokaula Extension	1
DUO Layout	1
CQAL Layout	1
Jagajyothi layout	1
Krishna Reddy Layout	1
Geetanjali Layout	1
Nirman Layout	1
Kamdhenu Nagar	1
Manonarayanapalya	1
BEML Layout 5th Stage	1
2nd phase jp nagar, jp nagar	1
Subramanya Nagar	1
KHB Colony Extension	1
Billapura	1
RR Layout	1
Nrupathunga Nagar	1

Muthyala Nagar	1
Vikram Nagar	1
Chokkahalli	1
TR Mill Road, Chamarajpet	1
T G extension	1
Chikka Gowdanapalya.	1
High grounds	1
Vittal Mallya Road	1
BTM Layout 1stage 9th Main	1
Whietfield,	1
RMV Extension Stage 2	1
Veer Sandra	1
Doddabele	1
Venkateshpuram	1
rr nagar	1
Gandhi Nagar	1
GKW Layout	1
anjananager magdi road	1
Ananthanagar Phase 1,Electronic City , phase 2	1
Lal Bahadur Shastri Nagar	1
Banashankari stage 2	1
Okalipura	1
Old Town	1
Vasantapura main road	1
Bapuji Layout	1
1st Stage Radha Krishna Layout	1
Lakshmiamma Garden	1
Dwarka Nagar	1
Ramaswamy Palya - Kammanahalli Main Road	1
BEML Layout 5th stage	1
Kannur	1
singapura paradise	1
Uvce Layout	1
CQAL LAYOUT C BLOCK	1
Badrappa Layout	1
manyata tech park	1
West of Chord Road	1
Prasanth Extension	1
Hanumagiri	1
Karnataka Shabarimala	1
JP nagar 9th Phase,	1
1 Annasandrapalya	1
CHIKKATIRUPATHI	1
JP Nagar 8th Phase,	1
Vijaya Bank Colony	1
1st Stage Domlur	1
EPIP AREA, WHITEFIELD	1
Maruthi Extension	1
Vidyapeeta	1
Meenakshi Layout	1
S R Layout	1
Basvasamithi Layout Vidyaranyapura	1
Mailasandra	1
kanakapura main road	1
3rd Phase JP Nagar	1
t.c palya	1

```
Manganahalli                                          1
Housing Board Layout Vijay Nagar                      1
 Whitefield                                           1
K R C kothanur                                        1
1Channasandra                                         1
Vijayabank bank layout                                1
Double Road                                           1
Saptagiri Layout                                      1
Thanisandra main road                                 1
Near ullas theater                                    1
Basnashankari,6th stage,                              1
Milk Colony                                           1
Kodanda Reddy Layout                                  1
Jaladarsini Layout                                    1
Madanayakahalli                                       1
N R Layout                                            1
Subbannaiah Palya                                     1
whitefiled                                            1
Medi Agrahara                                         1
Shirdi Sai Nagar                                      1
Abshot Layout                                         1
Name: count, dtype: int64
--------------------------------
size
2 BHK         5177
3 BHK         3760
1 BHK          453
4 Bedroom      413
3 Bedroom      385
2 Bedroom      300
5 Bedroom      174
6 Bedroom      127
1 Bedroom       84
4 BHK           61
8 Bedroom       60
7 Bedroom       51
9 Bedroom       32
6 BHK           13
5 BHK           12
10 Bedroom       7
1 RK             6
7 BHK            5
8 BHK            3
11 Bedroom       2
19 BHK           1
14 BHK           1
18 Bedroom       1
Name: count, dtype: int64
--------------------------------
```

In [44]:
```
pd.reset_option('display.max_rows')
pd.reset_option('display.max_columns')
```

In [45]:
```
# Xử lý cho trường "area_type" -> Dùng One-Hot Encoding
dummies = pd.get_dummies(df8["area_type"], drop_first=True)
```

```
df8 = df8.join(dummies)
df8.drop("area_type", axis=1, inplace=True)
```

In [46]:
```
# Xử lý cho trường "availability"
df8["is_ready"] = (df8["availability"] == "Ready To Move").astype(int)
df8.drop("availability", axis=1, inplace=True)
```

In [47]:
```
# Xử lý cho trường "location" (Thay bằng mean price của khu vực đó)
location_mean = df8.groupby("location")["price"].mean()
df8["location_encoded"] = df8["location"].map(location_mean)
df8.drop("location", axis=1, inplace=True)
```

In [48]:
```
# Xử lý cho trường "size"
df8["bhk"] = df8["size"].str.split().str[0].astype(int)
df8.drop("size", axis=1, inplace=True)
```

In [49]:
```
df8.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11128 entries, 0 to 11127
Data columns (total 11 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   bath                  11128 non-null  float64
 1   balcony               11128 non-null  float64
 2   price                 11128 non-null  float64
 3   total_sqft_float      11128 non-null  float64
 4   bhk                   11128 non-null  int64
 5   price_per_sqft        11128 non-null  float64
 6   Carpet  Area          11128 non-null  bool
 7   Plot  Area            11128 non-null  bool
 8   Super built-up  Area  11128 non-null  bool
 9   is_ready              11128 non-null  int64
 10  location_encoded      11128 non-null  float64
dtypes: bool(3), float64(6), int64(2)
memory usage: 728.2 KB
```

4.1.5. Chuẩn hóa dữ liệu (Normalization)

In [50]:
```
df8.head()
```

Out[50]:

	bath	balcony	price	total_sqft_float	bhk	price_per_sqft	Carpet Area	Plot Area	Super built-up Area	is_read
0	2.0	1.0	39.07	1056.0	2	3699.810606	False	False	True	
1	2.0	3.0	62.00	1440.0	3	4305.555556	False	False	False	
2	3.0	1.0	95.00	1521.0	3	6245.890861	False	False	True	
3	2.0	1.0	51.00	1200.0	2	4250.000000	False	False	True	
4	2.0	1.0	38.00	1170.0	2	3247.863248	False	False	True	

In [51]:
```
from sklearn.preprocessing import MinMaxScaler, StandardScaler
scaler = MinMaxScaler()
new_data_1 = scaler.fit_transform(df8)
standard = StandardScaler()
new_data_2 = standard.fit_transform(df8)
```

In [52]:
```
new_data_1
```

Out[52]:
```
array([[0.05882353, 0.33333333, 0.02714489, ..., 1.        , 0.        ,
        0.0592116 ],
       [0.05882353, 1.        , 0.04893112, ..., 0.        , 1.        ,
        0.08934912],
       [0.11764706, 0.33333333, 0.08028504, ..., 1.        , 1.        ,
        0.1968365 ],
       ...,
       [0.11764706, 1.        , 0.09643705, ..., 1.        , 1.        ,
        0.19052441],
       [0.05882353, 0.33333333, 0.04703088, ..., 0.        , 1.        ,
        0.0891749 ],
       [0.        , 0.33333333, 0.00617577, ..., 1.        , 1.        ,
        0.05783333]], shape=(11128, 11))
```

4.2. Giảm chiều dữ liệu

In [53]:
```
# PCA (Principal Component Analysis)
from sklearn.decomposition import PCA
pca = PCA(n_components=2)
X_pca = pca.fit_transform(new_data_1)
print(X_pca.shape)

# SVD (Singular Value Decomposition)
from sklearn.decomposition import TruncatedSVD
svd = TruncatedSVD(n_components=4)
X_svd = svd.fit_transform(new_data_2)
print(X_svd.shape)
```

```
(11128, 2)
(11128, 4)
```

Bài tập bổ sung (tự làm)

> Phần bài tập này là các câu hỏi mở rộng, làm tiếp theo bài toán ở trên. Sinh viên cần viết mã để thực hiện các yêu cầu dưới đây:

Bài tập 1: Sử dụng sns.boxplot() để quan sát đặc điểm phân bố dữ liệu của các trường số, mỗi trường này có outlier ko?

In [ ]:
```
# Sử dụng boxplot để quan sát phân bố của dữ liệu, phát hiện ngoại lai (xử lí nếu c
# Gợi ý: sns.boxplot(data_field)
vars = ['price', 'total_sqft_float', 'price_per_sqft', 'balcony', 'bath', 'bhk']
plt.figure(figsize=(16,12))

#Code ở đây
```

Bài tập 2: Viết hàm bỏ đi các điểm dữ liệu có price per sqft dựa trên mean, std của các ngôi nhà dựa trên từng vị trí

> Gợi ý: Xét trên từng vị trí (location), ngôi nhà thỏa mãn phải có $price_per_sqft \in [mean - std, mean + std]$

In [ ]:
```
def remove_pps_outliers(df):
  #Code ở đây
#--------
df9 = remove_pps_outliers(df8)
df9.shape
```

Bài tập 3: Loại bỏ outlier xét theo trường bkh (số phòng)

Xét theo từng khu vực địa lí và theo từng loại nhà với số lượng phòng khác nhau, có một số ngôi nhà có giá không hợp lí (outliers), hãy tìm cách loại bỏ các outlier này. Cần ghi rõ quy tắc ghi nhận outlier.

Gợi ý: Loại bỏ các căn nhà có giá trên mỗi sqft thấp hơn trung bình của nhóm cùng số phòng và cùng location (khi đủ dữ liệu). Với mỗi location:

- Tính mean price_per_sqft theo từng bhk
- Nếu có đủ dữ liệu (>5 căn): Loại bỏ những căn có giá/sqft thấp hơn mean

In [ ]:
```
#location_df.groupby('bhk')
```

In [ ]:
```
def remove_bhk_outliers(df):
  # Code ở đây

df10 = remove_bhk_outliers(df9)
df10.shape
```

Bài tập 4: Loại bỏ outlier khi xét trường 'bathroom'

In [ ]:
```
df10.bath.unique() #Có thể quan sát thấy một số căn nhà có số phòng tắm quá lớn (VD
```

In [ ]:
```
df10[df10.bath > df10.bhk+2]
```

In [ ]:
```
df11 = #Code ở đây, sao cho: df10[df10.bath < df10.bhk+2]
df11.shape
```

In [ ]:
```
df11.head()
# Quan sát lại kết quả sau khi xử lí với boxplot
```

Bài tập 5: Thực hiện các câu lệnh để trả lời các câu hỏi dưới đây:

- Thống kê giá nhà theo từng loại khu vực (area_type). Làm với df9:
- xem xét theo từng khu vực, thì giá nhà trung bình (price_per_sqft) là bao nhiêu, tương quan về giá nhà trung bình giữa các khu vực

Gợi ý: Phần này có thể đưa ra kết quả dạng bảng hoặc biểu đồ (cột, histogram, ...).

- Sử dụng các lệnh: df.groupby(), df.sortvalues(), ... để trích xuất giá trị
- Sử dụng matplotlib: plt.bar(), ...

In [ ]:
```
# Code ở đây
```